

Project Title: Duke Bike – E-Commerce Database Management System

Skill: DBMS – PAYMENT MANAGEMENT

1. Database Design

1.1 Payment Table

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Order_ID	INT	FK	References Orders
Payment_Method	VARCHAR(30)	-	UPI / Card / Net Banking
Transaction_ID	VARCHAR(100)	-	Transaction ID
Payment_Status	VARCHAR(20)	-	Successful / Failed
Payment_Date	DATE	-	Date of payment
Amount	DECIMAL(10,2)	-	Payment amount (in INR)

2. Table Relationships

Customer – Orders = 1 : M

Orders – Payment = 1 : 1

Orders – Order_Items = 1 : M

Product – Order_Items = 1 : M

Category – Product = 1 : M

3. Table Creation

3.1 Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER(5) PRIMARY KEY,  
3     Order_ID NUMBER(5) NOT NULL,  
4     Payment_Method VARCHAR2(30) NOT NULL,  
5     Transaction_ID VARCHAR2(100) NOT NULL,  
6     Payment_Status VARCHAR2(20) NOT NULL,  
7     Payment_Date DATE NOT NULL,  
8     Amount NUMBER(10,2) NOT NULL,  
9     CONSTRAINT fk_payment_order  
10    FOREIGN KEY (Order_ID)  
11    REFERENCES Orders(Order_ID)  
12 );
```

Table created.

4. Payment Data Insertion

```
SQL> INSERT INTO Payment  
2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,  
3  Payment_Status, Payment_Date, Amount)  
4 VALUES  
5 (701, 501, 'UPI', 'UPI5012026',  
6  'Successful', TO_DATE('01-OCT-2026', 'DD-MON-YYYY'), 10000.00);
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3   Payment_Status, Payment_Date, Amount)
  4 VALUES
  5 (702, 502, 'Card', 'CARD5022026',
  6   'Successful', TO_DATE('03-OCT-2026','DD-MON-YYYY'), 352000.00);

1 row created.
```

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3   Payment_Status, Payment_Date, Amount)
  4 VALUES
  5 (703, 503, 'UPI', 'UPI5032026',
  6   'Successful', TO_DATE('02-OCT-2026','DD-MON-YYYY'), 4599.00);

1 row created.
```

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3   Payment_Status, Payment_Date, Amount)
  4 VALUES
  5 (704, 504, 'Card', 'CARD5042026',
  6   'Successful', TO_DATE('04-OCT-2026','DD-MON-YYYY'), 225500.00);

1 row created.
```

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3   Payment_Status, Payment_Date, Amount)
  4 VALUES
  5 (705, 505, 'Net Banking', 'NB5052026',
  6   'Successful', TO_DATE('05-OCT-2026','DD-MON-YYYY'), 3299.50);

1 row created.
```

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Method, Transaction_ID,
  3   Payment_Status, Payment_Date, Amount)
  4 VALUES
  5 (706, 506, 'UPI', 'UPI5062026',
  6   'Failed', TO_DATE('06-OCT-2026','DD-MON-YYYY'), 1850.75);

1 row created.
```

```
SQL> COMMIT;

Commit complete.
```

5. Manage Payment Transactions

Display settings used for all query outputs:

```
SQL> SET LINESIZE 200
SQL> COLUMN Payment_Method FORMAT A14
SQL> COLUMN Transaction_ID FORMAT A14
SQL> COLUMN Payment_Status FORMAT A14
```

5.1 Display Successful Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
701	501	UPI	UPI5012026	Successful	01-OCT-26	10000
702	502	Card	CARD5022026	Successful	03-OCT-26	352000
703	503	UPI	UPI5032026	Successful	02-OCT-26	4599
704	504	Card	CARD5042026	Successful	04-OCT-26	225500
705	505	Net Banking	NB5052026	Successful	05-OCT-26	3299.5

5.2 Display Failed Payments

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
706	506	UPI	UPI5062026	Failed	06-OCT-26	1850.75

6. Modify Payment Status

Payment 706 (UPI) was retried by the customer and confirmed by the bank, so its status is changed from Failed to Successful.

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 706;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 706;
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_STATUS	PAYMENT_D	AMOUNT
706	506	UPI	UPI5062026	Successful	06-OCT-26	1850.75

7. Analyze Payment Methods

7.1 Calculate Total Transactions by Payment Method

```
SQL> SELECT
  2 Payment_Method,
  3 COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Method
  6 ORDER BY Payment_Method;
```

PAYMENT_METHOD	TOTAL_TRANSACTIONS
Card	2
Net Banking	1
UPI	3

7.2 Calculate Amount Collected by Payment Method

```
SQL> SELECT
  2     Payment_Method,
  3     SUM(Amount) AS Total_Amount
  4 FROM Payment
  5 WHERE Payment_Status = 'Successful'
  6 GROUP BY Payment_Method
  7 ORDER BY Payment_Method;
```

```
PAYMENT_METHOD TOTAL_AMOUNT
-----
Card            577500
Net Banking     3299.5
UPI             16449.75
```

8. Payment Transaction Report

```
SQL> SET LINESIZE 220
SQL> COLUMN Customer_Name FORMAT A15
SQL> COLUMN Product_Name FORMAT A24
SQL> COLUMN Category_Name FORMAT A13
SQL> COLUMN Payment_Method FORMAT A14
SQL> COLUMN Transaction_ID FORMAT A14
SQL> COLUMN Payment_Status FORMAT A14
```

```
SQL> SELECT
 2      p.Payment_ID,
 3      o.Order_ID,
 4      c.Customer_ID,
 5      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
 6      pr.Product_ID,
 7      pr.Product_Name,
 8      cat.Category_ID,
 9      cat.Category_Name,
10      p.Payment_Method,
11      p.Transaction_ID,
12      p.Payment_Date,
13      p.Amount,
14      p.Payment_Status
15 FROM Payment p
16 JOIN Orders o
17   ON p.Order_ID = o.Order_ID
18 JOIN Customer c
19   ON o.Customer_ID = c.Customer_ID
20 JOIN Order_Items oi
21   ON o.Order_ID = oi.Order_ID
22 JOIN Product pr
23   ON oi.Product_ID = pr.Product_ID
24 JOIN Category cat
25   ON pr.Category_ID = cat.Category_ID
26 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID	CUSTOMER_NAME	PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	CATEGORY_NAME	PAYMENT_METHOD	TRANSACTION_ID	PAYMENT_D	AMOUNT	PAYMENT_STATUS
706	506	106	Sneha Iyer	406	Crash Guard Set	303	Accessories	UPI	UPI5062026	06-OCT-26	1850.75	Successful
705	505	105	Rahul Nair	405	Full-Face Helmet	302	Riding Gear	Net Banking	NB5052026	05-OCT-26	3299.5	Successful
704	504	104	Divya Menon	404	Duke 250	301	Motorcycles	Card	CARD5042026	04-OCT-26	225500	Successful
702	502	102	Priya Ravi	402	Duke 390	301	Motorcycles	Card	CARD5022026	03-OCT-26	352000	Successful
703	503	103	Karthik Raja	403	Riding Jacket	302	Riding Gear	UPI	UPI5032026	02-OCT-26	4599	Successful
701	501	101	Arjun Kumar	401	Duke 200 Booking Amount	301	Motorcycles	UPI	UPI5012026	01-OCT-26	10000	Successful

Appendix A: Supporting Tables (run before Section 3)

```
CREATE TABLE Customer (
  Customer_ID NUMBER(5) PRIMARY KEY,
  First_Name VARCHAR2(50) NOT NULL,
  Last_Name VARCHAR2(50) NOT NULL,
  Email VARCHAR2(100) UNIQUE
);
CREATE TABLE Category (
  Category_ID NUMBER(5) PRIMARY KEY,
```

```

    Category_Name VARCHAR2(50) NOT NULL
);
CREATE TABLE Product (
    Product_ID NUMBER(5) PRIMARY KEY,
    Product_Name VARCHAR2(100) NOT NULL,
    Category_ID NUMBER(5) NOT NULL,
    Price NUMBER(10,2) NOT NULL,
    CONSTRAINT fk_product_category FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
);
CREATE TABLE Orders (
    Order_ID NUMBER(5) PRIMARY KEY,
    Customer_ID NUMBER(5) NOT NULL,
    Order_Date DATE NOT NULL,
    Total_Amount NUMBER(10,2) NOT NULL,
    CONSTRAINT fk_orders_customer FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
CREATE TABLE Order_Items (
    Order_Item_ID NUMBER(5) PRIMARY KEY,
    Order_ID NUMBER(5) NOT NULL,
    Product_ID NUMBER(5) NOT NULL,
    Quantity NUMBER(3) NOT NULL,
    Unit_Price NUMBER(10,2) NOT NULL,
    CONSTRAINT fk_items_order FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID),
    CONSTRAINT fk_items_product FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)
);

INSERT INTO Customer VALUES (101, 'Arjun', 'Kumar', 'arjun.kumar@example.com');
INSERT INTO Customer VALUES (102, 'Priya', 'Ravi', 'priya.ravi@example.com');
INSERT INTO Customer VALUES (103, 'Karthik', 'Raja', 'karthik.raja@example.com');
INSERT INTO Customer VALUES (104, 'Divya', 'Menon', 'divya.menon@example.com');
INSERT INTO Customer VALUES (105, 'Rahul', 'Nair', 'rahul.nair@example.com');
INSERT INTO Customer VALUES (106, 'Sneha', 'Iyer', 'sneha.iyer@example.com');

INSERT INTO Category VALUES (301, 'Motorcycles');
INSERT INTO Category VALUES (302, 'Riding Gear');
INSERT INTO Category VALUES (303, 'Accessories');

INSERT INTO Product VALUES (401, 'Duke 200 Booking Amount', 301, 10000.00);
INSERT INTO Product VALUES (402, 'Duke 390', 301, 352000.00);
INSERT INTO Product VALUES (403, 'Riding Jacket', 302, 4599.00);
INSERT INTO Product VALUES (404, 'Duke 250', 301, 225500.00);
INSERT INTO Product VALUES (405, 'Full-Face Helmet', 302, 3299.50);
INSERT INTO Product VALUES (406, 'Crash Guard Set', 303, 1850.75);

INSERT INTO Orders VALUES (501, 101, TO_DATE('30-SEP-2026','DD-MON-YYYY'), 10000.00);
INSERT INTO Orders VALUES (502, 102, TO_DATE('02-OCT-2026','DD-MON-YYYY'), 352000.00);
INSERT INTO Orders VALUES (503, 103, TO_DATE('01-OCT-2026','DD-MON-YYYY'), 4599.00);
INSERT INTO Orders VALUES (504, 104, TO_DATE('03-OCT-2026','DD-MON-YYYY'), 225500.00);
INSERT INTO Orders VALUES (505, 105, TO_DATE('04-OCT-2026','DD-MON-YYYY'), 3299.50);
INSERT INTO Orders VALUES (506, 106, TO_DATE('05-OCT-2026','DD-MON-YYYY'), 1850.75);

INSERT INTO Order_Items VALUES (601, 501, 401, 1, 10000.00);
INSERT INTO Order_Items VALUES (602, 502, 402, 1, 352000.00);

```

```
INSERT INTO Order_Items VALUES (603, 503, 403, 1, 4599.00);  
INSERT INTO Order_Items VALUES (604, 504, 404, 1, 225500.00);  
INSERT INTO Order_Items VALUES (605, 505, 405, 1, 3299.50);  
INSERT INTO Order_Items VALUES (606, 506, 406, 1, 1850.75);  
  
COMMIT;
```